Greecon
SH.P.K

03

HARDWARE SHEET HW03 · QUALIFIED BY SPECIFICATION

The data cable and labels

The two-wire cable every meter, module and sensor shares, and the labels that let the next person understand the kit.

QUALIFIES	Qualified by specification. Shielded twisted pair for the RS-485 line, ordinary network patch cords, ferrules and labels. How it is run matters more than what is bought.
USED IN	Kits H, S, F and W (H hotel or guesthouse, S solar plant, F farm or greenhouse, W water tank and pump station)
FOR	The project manager who runs the cable, and the electrician who connects it inside a board
COVERS	How it qualifies · what it does at each kind of site · settings · fitting · in the platform · the acceptance test · what goes wrong · what is not yet proven · the buying record



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HW03

The data cable and labels

Every meter, module and field sensor of a kit hangs on one two-wire data cable (RS-485) that starts at the converter. Almost every fault that looks random comes from how this cable was run, not from what was bought, so this sheet is mostly rules. The labels are on the same sheet because they are fitted at the same time, and because the next person to open the enclosure has nothing else to go by.

At a glance

WHAT IT DOES	The RS-485 data cable (shielded twisted pair), network patch cords, ferrules and labels.
HOW IT QUALIFIES	**Qualified by specification.** Shielded twisted pair for the RS-485 line, ordinary network patch cords, ferrules and labels. How it is run matters more than what is bought.
SPECIFICATION	Shielded twisted pair for the RS-485 line, network patch cords, ferrules, labels
FITTED BY	The project manager runs and labels it. The electrician connects its ends at any device that sits inside a distribution board.
PLANNING ESTIMATE (EUR)	40 to 80. Not a quote.
USED IN	Kits H, S, F and W. The table under *At each kind of site* says what it does at each.

How it qualifies

It qualifies if:

- ☐ The data cable is **shielded twisted pair**: one pair for A and B, with a conductor for the ground where the devices have a GND terminal.
- ☐ The network cables are ordinary patch cords, one for the box and one for the converter, tested with a cable tester.
- ☐ There are ferrules for every stranded end, and labels for every device and for both ends of every wire.

STOP

Do not use unshielded cable for the data line, do not run it in the same duct as power cables, and do not join devices with branches. A cable that qualifies and is run as a star still fails.

The rules of the cable

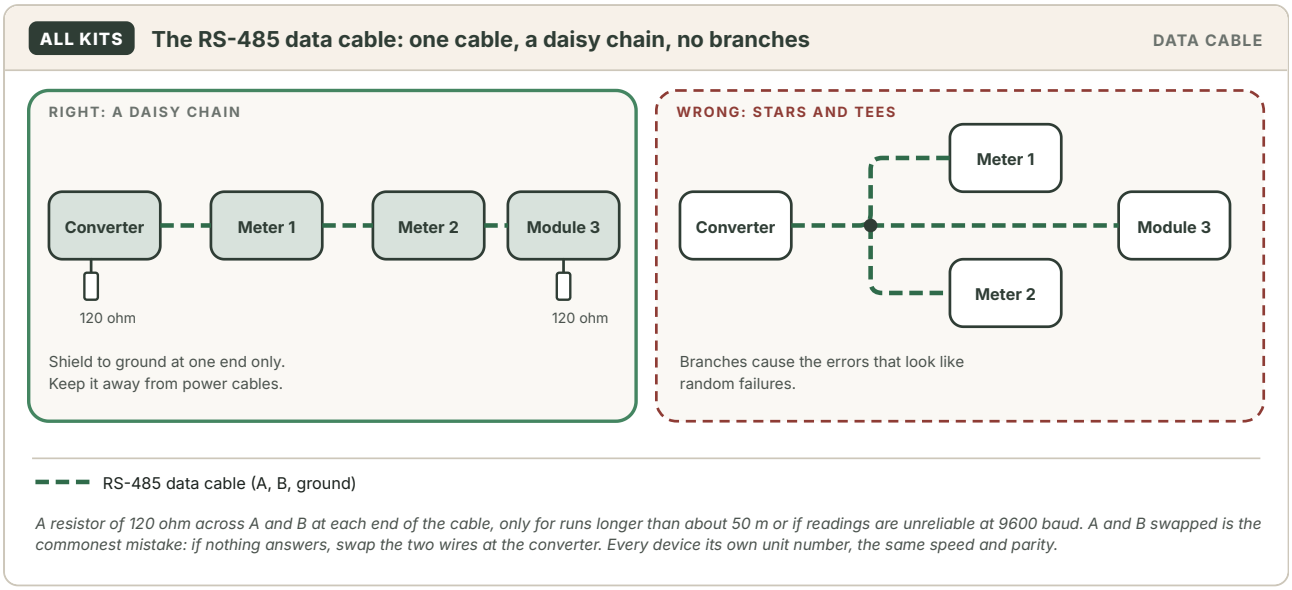


Figure 4-1. The RS-485 data cable: one cable and a daisy chain, never branches.

- **One cable, a daisy chain:** from the converter to the first device, on to the next, and so on. No branches: stars and tees cause the errors that look like random failures.
- **A to A and B to B at every device.** A and B swapped is the commonest mistake. If nothing answers, swap the two wires at the converter.
- **The shield joined to ground at one end only.** Keep the cable away from power cables, and note any place where it has to cross one.
- **Every device its own number, the same speed.** They are set at the office: on site you check each label and do not change it.
- **A 120 ohm resistor** across A and B at each end of the cable only for a run longer than about 50 metres, or if readings are unreliable at 9600 baud.
- **Plan the whole run under 100 metres.** A sensor further away than that needs another plan: ask the founder before promising it.

At each kind of site

Kind of site	In its kit	How many	What it does there
H Hotel or guesthouse	IN THE KIT	1 lot	One run from the converter through the grid meter and the solar meter (in the board, where the electrician connects it) and the water module to the pump contact module.
S Solar plant with grid connection	IN THE KIT	1 lot	One run from the converter through the grid meter, the solar meter and, if fitted, the load meter. Every end at a meter is the electrician's.
F Farm or greenhouse with irrigation	IN THE KIT	1 lot	The longest run of the four: from the converter in the pump house through the soil probe and the weather sensor in the field and the water module to the pump contact module. Plan it under 100 metres.
W Water tank and pump station	IN THE KIT	1 lot	A short run inside the pump house: from the converter through the pump meter (if fitted) and the water module to the pump contact module.

Labels

Label, on the day, and photograph before the enclosure is closed:

- [ ] The unit number on every device (it was set at the office), and the settings on the converter.
- [ ] The number of every wire, at both ends.
- [ ] The site name and the date, on a sticker inside the lid of the enclosure, and on the box.

Never write a code or a password on anything.

Fitting

PROJECT MANAGER runs the cable, and connects it at the converter, the modules and the field sensors. ELECTRICIAN connects its ends at any device inside a distribution board (the meters), because their terminals are inside the board.

- [ ] Walk the route first, and write down its length.
- [ ] Strip, fit ferrules, and connect A, B and the ground in the same order at every device.
- [ ] Join the shield to ground at the converter's end only.
- [ ] Fix the cable so that it cannot be pulled, and seal the glands.

In the platform

The cable is not in the platform. What it carries is: when it is right, **Find equipment** finds every device at its own number, and step 6 of the setup list, **Every device answers the box**, turns **Done** by itself.

Acceptance test

- [ ] The patch cords pass the cable tester.
- [ ] **Find equipment** finds every device of the kit, each at its own number.
- [ ] On the box, `greecon-edge status` lists every device as **ok**, and still does an hour later.
- [ ] The length of the run, and whether a resistor was fitted, are written on the commissioning record.

What goes wrong

What you see	Likely cause	What to do
Nothing behind the converter answers	A and B swapped	Swap the two wires at the converter
One device does not answer, the others do	That device's own A and B, its power, or the number on its label	Check its two wires and its power. Read its label against the install card. Numbers are not changed on site
Devices answer, then fail now and then	A branch in the cable, the shield joined to ground at both ends, a run beside power cables, or a long run with no resistor	Make it one line. Ground the shield at one end. Move it away from power. Over about 50 metres, fit the 120 ohm resistors
Two devices answer over each other	Two devices with the same number: one was not numbered at the office	That part goes back to the office, or the spare that was numbered there is fitted

Not yet proven

- The figures on this sheet (under 100 metres, a resistor over about 50 metres) are rules of thumb from the hardware guide, not measurements at a Greecon site. Write down the length of every run and whether a resistor was needed: the first sites will say whether the rule holds.

Buying record

Ask three suppliers for a quote and keep the quotes in the shared drive. Buy one, pass the acceptance test with it, and only then buy more. For every part that is bought the hardware register keeps the make and model, the supplier, the date, the serial number or the batch, and the invoice.

	Quote 1	Quote 2	Quote 3
Supplier			
Make and model offered			
Price (EUR, without VAT), and delivery			
Meets every line of *It qualifies if*	☐ Yes	☐ Yes	☐ Yes

Bought	Answer
Make, model and version	
Supplier, date and invoice	
Serial number or batch	
Acceptance test passed on (date), by	